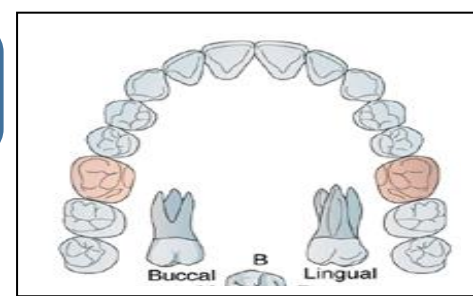


Maxillary Molars

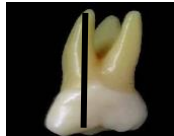


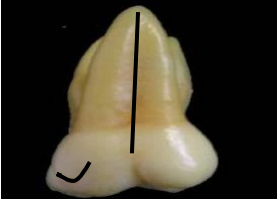
- They are 3 in each quadrant .
- Each molar has 3 roots (mesiobuccal- distobuccal – and palatal root), the palatal root is the longest.
- Roots are widely divergent ?? to give maximum anchorage against forces.
- Each molar has five surfaces.
- Thickness > width
- All Maxillary molars have lingual convergence except upper first molar .
- All maxillary molars have distal convergence .

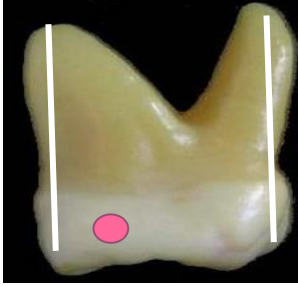
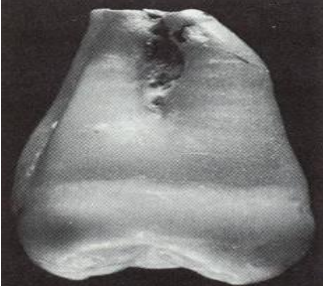
- As we go posteriorly : ↑
- As we go posteriorly : ↓

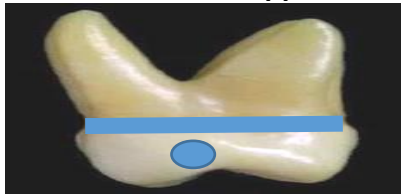
1 (7y)	2 (8y)	3 (10-12y)	4 (10-11y)	5 (10-12y)	6 (6y)	7 (12-13y)	8 (18y)
1 (6y)	2 (7y)	3 (9-10y)	4 (10-12y)	5 (11-12y)	6 (6y)	7 (11-12y)	8 (18y)

	Beginning of calc.	Crown completed	Eruption	Root completed
<u>6</u> 6	At birth At birth	-3 years	6-7 y 6-7 y.	+3 years
<u>7</u> 7	2½-3 y. 2½-3 y.	-5 years	12 y. 12 y.	
<u>8</u> 8	7-9 y. 8-10 y.		18 y. 18 y.	

Buccal Aspect	Upper 6		Upper 7		Upper 8	
Geometric outline	Trapezoidal with small uneven side present cervically.					
Mesial outline	Nearly straight till the contact area					
MCA	At the junction of the occlusal and middle 1/3					
Distal outline	Convex till the contact area					
DCA	At the middle 1/3				No DCA	
Occlusal outline:	Mesio-buccal cusp is broad and short. Disto-buccal cusp narrow, long and sharp. N.B.: The lingual cusps could be seen from this aspect due to the rhomboid occlusal outline.		In <u>7</u> MB cusp is larger and longer than DB cusp.		MB cusp is wider and longer than DB cusp.	
Cervical outline	Slightly convex toward the root.					
Anatomical landmarks	*Convex buccal surface. *Cervical ridge. *Slight depression occlusal to the cervical ridge. * Buccal ridge on each buccal cusp. *Buccal groove may terminate by buccal pit.					
Roots	* Root trunk: 4 millimeters *3 roots are seen . *MBR curves mesially from the CL till M 1\3, then D&ends in a blunt apex which is in line with MBC. *DBR is straighter & tends to curve mesially as it approaches the apical 1\3. * Deep developmental depression extends on root trunk. * The 2 buccal roots are nearly equal in length but the palatal root is the longest. *Roots are not confined to the outline		-The root trunk is longer in <u>7</u> . -The roots are nearly parallel &more inclined distally. -MB root apex is in line with buccal groove. - roots are within the outline		-The roots are often fused making one large short root. -Roots are within the outline	
Notes	- It is the largest tooth in the maxillary arch. -Buccal convergence		Smaller in length		Smaller in length	

Lingual Aspect	Upper 6	Upper 7	Upper 8
			
N.B	* No lingual convergence		Lingual convergence
Mesial outline	Nearly straight and form with the mesial slope of the ML cusp.		
Distal outline	Convex and form semicircle with the distal slope of the DL cusp.		
Occlusal outline	-ML cusp is the largest and longest cusp (3/5 of the crown width). -DL cusp is spheroidal.		* DL cusp is smaller in <u>Z</u> . It may be missed (3-cusp type) Usually one large lingual cusp without lingual groove.
Cervical outline	Irregular and slightly convex toward the root.		
Anatomical landmarks:	*Convex lingual surface with maximum convexity at middle third. * Tubercle of Carabelli on ML cusp (60%). Outlined by a faint DG (the 5 th cusp D groove). *Lingual developmental groove runs cervically till the center of the crown. *Apical to the lingual DG; A shallow depression extends till the middle 1/3 of the palatal root.		No tubercle of Carabelli
The Roots	-3 roots r seen, the palatal root is the widest root from this aspect. -Parts of the buccal roots are seen behind the lingual one. -Its apex is in line with the lingual groove		-The apex of the lingual root is in line with the DLC. -There is no developmental depression on the palatal root.

Mesial aspect	Upper 6 	Upper 7 	Upper 8 
The geometric outline	Trapezoidal with small uneven side occlusally مقلوب		
Buccal outline	Convex at cervical 1/3 denoting cervical ridge. * Concave at the middle 1/3. * Convex at the occlusal 1/3 representing circumscribed MB cusp.		
Lingual outline	Convex with the crest of curvature at the middle 1/3. * The lingual outline dips inward to illustrate the tubercle.		
Occlusal outline	* Represented by ML, MB cusps * Irregular MMR which curves cervically.		
Cervical outline	Irregular and convex occlusally		
Anatomical landmarks	Mesial contact area at the junction between middle and occlusal 1/3, buccal to the midline.	* No tubercle of Carabelli in <u>7</u> .	
The Roots	2 roots are seen, MB & palatal. MB root occupies 2/3 of the BL dimension. It extends upwards and outwards. Palatal root is long and narrow; it is banana shaped extending beyond the crown confines. It has a convex lingual outline extending lingually till the M 1/3 then curves buccally. Root trunk = 3 mm.	* MB and P roots in <u>7</u> are within confines of the BL outlines of the crown.	Characterized by fusion of the roots and the bifurcation in the apical 1/3.
Distal Aspect		* Distal Convergence. ** Convex distal surface. *** Distal MR curved cervically.	Convex surface with no contact area. * The distal surface short and narrow. So most of the occlusal surface is seen.

DISTAL ASPECT of upper 6

*Distal Convergence.

*Convex distal surface.

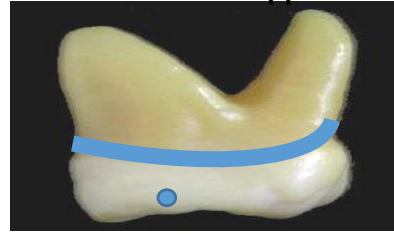
*Distal MR curved cervically exposing parts of the 4 cusps.

*Straight cervical line.

*Contact area more cervically, broader.

*Root trunk= 5mm.

*3 roots can be seen (the MB root is the widest while the P root is the longest).

MESIAL ASPECT of upper 6

* Wider mesial surface.

* Flat mesial surface.

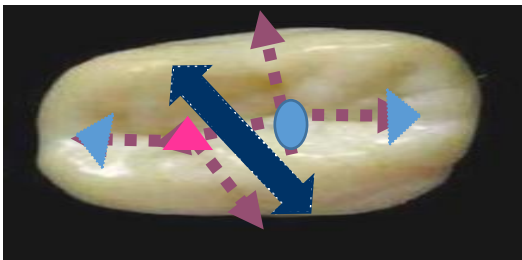
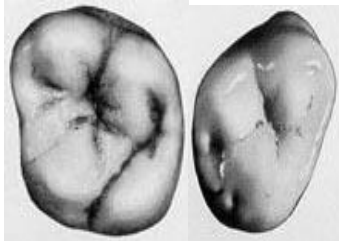
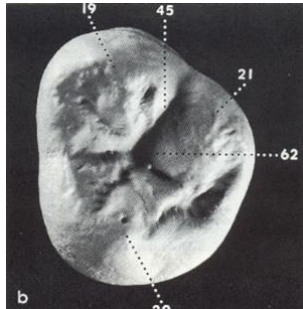
* Mesial marginal ridge less curved cervically.

*Cervical line convex occlusally.

*Contact area more occlusally.

*Root trunk = 3mm.

*2 roots can be seen with the same relations.

Occlusal Aspect	Upper 6	Upper 7		Upper 8	
					
		4cusp type 3 cusp Type		4 cusp type	3 cusp type
The geometric outline	Rhomboidal. BL is greater than MD dimension.	Rhomboidal (more frequent)	Heart shape (less frequent)	Same as 6	Same as 7 (3 cusp type)
Elevations	*4 cusps with 4 triangular ridges (ML>MB>DL>DB) and tubercle. *Oblique ridge between ML,DB triangular ridge. * MMR and DMR.			Same as 6	Same as 7 (3cusp type)
Depressions	*2 major fossae (<u>central which is triangular and distal which is linear</u>). *2 Minor fossae (mesial and distal triangular fossae). *Developmental grooves (Buccal, Central (from the central fossa to the M triangular fossa), transverse groove of oblique ridge (from the central fossa to the distal fossa), Lingual groove and the 5 th cusp groove).			*numerous supplemental grooves*	
Note	* Disto-buccal convergence. * ML, DB angles are obtuse. *MB, DL angles are acute.	*(obtuse angles r more obtuse & acute r more acute). *Disto-lingual convergence. *No tubercle of Carabelli.	*Lingual convergence. *LC &DBC r joined by oblique ridge. *No disto-lingual cusp and no lingual groove.	There are disto- lingual cusp, lingual groove and oblique ridge.	Lingual convergence, no lingual groove and no oblique ridge. No tubercle of Carabelli.

Notes:

- 1- The occlusal surface of 8 has numerous supplemental grooves.

Variations of upper 8

- So large and malformed.
- Underdeveloped and conical with small tapering root.
- Accessory cusps and roots.

Pulp cavity of all maxillary molars :

- The pulp chamber is broader Bucco-lingually than mesio-distally.
- There is pulp horn beneath each cusp. (4 or 3 pulp horns according to the cusp type)
- The floor is apical to the cervical line.

► Root Canals:

- 3 main root canals. (MB- DB-P)
- A fourth canal is usually present in the MB root.

Note:

Cervical cross section: Rhomboidal.

Mid root section: MB root canal (Oval- Kidney).

DB root canal (Round-Oval).

P root canal (Round- Oval).